

Case name: EngineK

Description:

Thu Jan 1 23:26:09 2026

Table 1. Propellant Specification

Component	Temperature, K	Pressure, MPa	Enthalpy, kJ/mol	Enthalpy, kJ/kg	Mass fraction
C ₂ H ₈ N ₂ (L),UDMH	298.1	0.1013	48.9001	813.668	0.1562500
N ₂ H ₄ (L)	298.1	0.1013	50.3798	1572.150	0.1562500
N ₂ O ₄ (L)	298.1	0.1013	-17.5489	-190.726	0.6875000
Total			16.1670	241.659	

Propellant exploded formula:

C_{0.347865} H_{2.696253} N_{2.000000} O_{1.999478} (based on 1 mole)

C_{4.938747} H_{38.279499} N_{28.394586} O_{28.387168} (by mass %)

α : 0.9782929 (oxidizer excess coefficient)

O/F: 2.2000000

O/F₀: 2.2488153 (stoichiometric)

ρ : 1197.90819 kg/m³

Table 2. Combustion Properties

Parameter	Injector	Nozzle inlet	Nozzle throat	Nozzle exit Unit
Pressure	0.8481	0.8268	0.4852	0.0008 MPa
Temperature	3140.9746	3136.0463	2990.4841	1345.2307 K
Enthalpy	5503.8314	5170.3721	-8423.0849	-128130.9341 J/mol
	241.6557	226.9717	-365.1508	-5163.5414 kJ/kg
Entropy	271.0654	271.2206	274.6449	295.4466 J/(mol·K)
	11.9016	11.9062	11.9062	11.9062 kJ/(kg·K)
Internal energy	-20611.7140	-20904.1966	-33287.3814	-139315.8169 J/mol
	-904.9947	-917.6634	-1443.0477	-5614.2803 kJ/kg
Specific heat (p=const)	6.5221	6.5273	6.2871	1.7296 kJ/(kg·K)
Specific heat (V=const)	5.6123	5.6178	5.4431	1.3945 kJ/(kg·K)
Gamma	1.1621	1.1619	1.1551	1.2403
Isentropic exponent	1.1284	1.1282	1.1259	1.2403
Gas constant	0.3651	0.3650	0.3604	0.3351 kJ/(kg·K)
Molecular weight	22.7755	22.7798	23.0674	24.8145
Density	0.7396	0.7224	0.4501	0.0019 kg/m ³
Sonic velocity	1137.4760	1136.4098	1101.6411	747.6985 m/s
Mach number	0.0000	0.1508	1.0000	4.3974
Area ratio	4.0000	4.0000	1.0000	80.1620 A/At
Mass flux	123.7911	123.7911	495.8909	6.1861 kg/(m ² ·s)
Viscosity	1.0051	1.0040	0.9733	0.5630 x10 ⁻⁴ kg/(m·s)
Conductivity, frozen	0.2995	0.2991	0.2854	0.1257 W/(m·K)
Specific heat (p=const), frozen	2.0053	2.0049	1.9948	1.7194 kJ/(kg·K)
Prandtl number, frozen	0.6728	0.6730	0.6803	0.7703
Conductivity, effective	1.4143	1.4139	1.2872	0.1258 W/(m·K)
Specific heat (p=const), effective	6.5221	6.5273	6.2871	1.7203 kJ/(kg·K)
Prandtl number, effective	0.4635	0.4635	0.4754	0.7698

Table 3. Combustion Products

Product	Injector mass fraction	Injector mole fraction	Nozzle inlet mass fraction	Nozzle inlet mole fraction	Nozzle throat mass fraction	Nozzle throat mole fraction	Nozzle exit mass fraction	Nozzle exit mole fraction
CO	0.0759473	0.0617537	0.0758178	0.0616601	0.0684592	0.0563785	0.0061276	0.0054285
CO2	0.1095096	0.0566724	0.1097130	0.0567884	0.1212764	0.0635662	0.2192135	0.1236018
COOH	0.0000016	0.0000008	0.0000016	0.0000008	0.0000009	0.0000005		
H	0.0008336	0.0188356	0.0008318	0.0187981	0.0006580	0.0150592	0.0000001	0.0000025
H2	0.0046542	0.0525831	0.0046458	0.0524976	0.0040767	0.0466491	0.0008963	0.0110328
H2O	0.2930966	0.3705403	0.2932545	0.3708099	0.3032241	0.3882570	0.3550231	0.4890121
H2O2	0.0000059	0.0000039	0.0000058	0.0000039	0.0000036	0.0000024		
HCO	0.0000008	0.0000007	0.0000008	0.0000006	0.0000004	0.0000003		
HCOOH	0.0000001	0.0000001	0.0000001	0.0000001				
HNO	0.0000060	0.0000044	0.0000058	0.0000043	0.0000033	0.0000024		
HNO2	0.0000015	0.0000007	0.0000015	0.0000007	0.0000009	0.0000004		
HO2	0.0000598	0.0000413	0.0000590	0.0000407	0.0000382	0.0000267		
N	0.0000040	0.0000066	0.0000040	0.0000065	0.0000021	0.0000035		
N2	0.4103569	0.3336274	0.4104043	0.3337288	0.4120108	0.3392652	0.4187378	0.3709200
N2O	0.0000029	0.0000015	0.0000028	0.0000014	0.0000017	0.0000009		
NH	0.0000011	0.0000017	0.0000011	0.0000017	0.0000005	0.0000008		
NH2	0.0000004	0.0000006	0.0000004	0.0000006	0.0000002	0.0000003		
NH3	0.0000003	0.0000004	0.0000003	0.0000004	0.0000001	0.0000002		
NO	0.0179210	0.0136025	0.0178202	0.0135285	0.0143922	0.0110641		
NO2	0.0000158	0.0000078	0.0000156	0.0000077	0.0000102	0.0000051		
O	0.0073821	0.0105085	0.0073591	0.0104778	0.0056456	0.0081395		
O2	0.0407902	0.0290327	0.0407708	0.0290244	0.0371849	0.0268058		
OH	0.0394081	0.0527734	0.0392839	0.0526169	0.0330098	0.0447717	0.0000015	0.0000021
Gaseous fraction:	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Condensed fraction:	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000

Table 4. Ideal Performance

Parameter	Sea level	Optimum expansion	Vacuum Units
Characteristic velocity		1688.8964	m/s
Effective exhaust velocity	-12954.4412	3287.9164	3425.0067 m/s
Specific impulse (by mass)	-12954.4412	3287.9164	3425.0067 N-s/kg
Specific impulse (by weight)	-1320.9854	335.2742	349.2535 s
Thrust coefficient	-7.6704	1.9468	2.0280
Thrust	-63.2744	16.0594	16.7290 kN
Altitude	0.0000	32.3188	km
Ambient pressure	1.0000	0.0084	0.0000 atm

Table 5. Estimated Delivered Performance

Parameter	Sea level	Optimum expansion	Vacuum Units
Characteristic velocity		1614.5968	m/s
Effective exhaust velocity	-12457.8037	3070.0045	3201.0638 m/s
Specific impulse (by mass)	-12457.8037	3070.0045	3201.0638 N-s/kg
Specific impulse (by weight)	-1270.3424	313.0533	326.4177 s
Thrust coefficient	-7.7157	1.9014	1.9826

Parameter	Sea level	Optimum expansion	Vacuum Units
Thrust	-60.8486	14.9951	15.6352 kN
Altitude	0.0000	32.3188	km
Ambient pressure	1.0000	0.0084	0.0000 atm

Table 6. Altitude Performance

Table 7. Throttled Performance

Table 8. Chamber Size

Combustion chamber size			Nozzle size		
Dc	218.99	mm	Type	Parabolic nozzle	
Dt	109.50	mm	Rn	20.91	mm
Lcyl	94.52	mm	Tn	37.64	deg
Lc	255.10	mm	Te	10.46	deg
L*	780.00	mm	De	980.35	mm
R1	82.12	mm	Le	1302.23	mm
R2	163.26	mm	Le/Dt	11.89	
b	30.00	deg	Le/Lc15	80.00	%
Ac/At	4.00		Ae/At	80.16	

Parameter		Engine	Chamber
Thrust	sea level	-60.8486	-60.8486 kN
	opt exp	14.9951	14.9951 kN
	vacuum	15.6352	15.6352 kN
Specific Impulse	sea level	-12457.8037	-12457.8037 N·s/kg
	opt exp	3070.0045	3070.0045 N·s/kg
	vacuun	3201.0638	3201.0638 N·s/kg
Mass flow rate	total	4.8844	4.8844 kg/s
	oxidizer	3.3580	3.3580 kg/s
	fuel	1.5264	1.5264 kg/s
	film coolant	n.a.	n.a. kg/s
Number of chambers		1	

Table 10. Thermal Analysis